

Test-Time Scaling of Vision-Language-Action Models via Self-Certainty

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Abstract: While generalist robot policies like Vision-Language-Action (VLA) models are able to handle a wide range of tasks, they are often not reliable and may fail at test time. To address this, we adapt the verifier-free, best-of-N selection strategy of self-certainty [1]—a model’s internal confidence score—for improving test-time performance of VLAs. By sampling multiple actions and selecting the most confident one, this strategy trades increased test-time computation for potential enhanced performance without any model retraining. We explore ways of defining and obtaining the self-certainty scores to make the strategy compatible with continuous action generation and diffusion/flow sampling. Experiments in both simulation and real world demonstrate that this “plug-and-play” approach yields significant success rate improvements over standard greedy decoding for autoregressive policies and random sampling for diffusion-based or flow-based policies, establishing it as a general framework for the test-time scaling of VLA models.

Keywords: Vision-Language-Action Models, Test-Time Scaling, Self-Certainty

1 Introduction

Recent advancements in robot learning, fueled by large-scale datasets, have produced a new generation of generalist robot policies [2, 3]. In particular, Vision-Language-Action (VLA) models have achieved remarkable success, showcasing strong generalization capabilities and excelling at a diverse range of complex tasks [4, 5, 6, 7, 8]. Despite these strides, the very design that makes these VLA models generalist—often relying on a reactive control loop—can also render them brittle at test time. They may fail at test time when faced with novel situations not comprehensively covered in the training data [9] or through compounding errors over long-horizon tasks.

To address this challenge, one promising direction, inspired by recent advances in Large Language Models (LLMs), is to shift focus from scaling with more data and model parameters to scaling test-time computation [10]. Following this strategy, recent works [11, 12] adopt a sample-and-verify approach: they first sample multiple candidate actions from the VLA policy, and then employ a trained verifier to select the optimal one. While promising, this methodology has a key limitation: the verifier itself can be data-intensive and computationally expensive to train, and its robustness may degrade when faced with out-of-distribution tasks. *Is it possible to rank actions proposed by a VLA model without an external verifier?*

In this work, we introduce a verifier-free method for test-time selection. We draw inspiration from a recent study on LLM reasoning [1], which introduces *self-certainty*, a distributional confidence metric that assesses the quality of an LLM’s reasoning trace. It is calculated by aggregating token-level statistics, e.g., KL-divergence and entropy, to filter out low-quality answers without an external reward model. Analogously, we adapt this concept to the robotics domain. We treat the whole action

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chunk of VLAs as a “policy trace”, and for each candidate chunk generated by the VLA model, we compute a self-certainty score to rank and select the most reliable action.

However, adapting the principle of self-certainty from the discrete domain of language to the continuous domain of robotic actions presents a significant challenge. LLMs generate text by predicting discrete tokens over a finite vocabulary, for which the original self-certainty metric is well-defined. In contrast, many state-of-the-art VLA policies model high-dimensional, continuous actions, often employing diffusion or flow-based methods that operate in a continuous space rather than over a discrete distribution [6, 13, 7, 8]. This fundamental difference in output modality makes the original formulation of self-certainty inapplicable. Therefore, a core contribution of this work is to propose a novel formulation of self-certainty specifically tailored for continuous actions, potentially generated by diffusion-based or flow-based policies.

In addition, we address the challenge of selecting the best candidate in a continuous action space. The Borda Voting mechanism proposed by Kang et al. [1] relies on grouping discrete, identical answers for voting. This approach is ill-suited for continuous VLA actions, where generating identical outputs is practically impossible. We therefore introduce a novel selection mechanism specifically for continuous spaces, which combines the self-certainty score of each candidate with a similarity-based agreement score, allowing for a robust selection without requiring discrete voting.

To validate the effectiveness and generality of our approach, we conduct extensive experiments in both simulation and the real world. A key advantage of our method is its plug-and-play nature, allowing it to be applied to a wide range of pretrained VLA models without any retraining. We demonstrate its utility across diverse model architectures and sizes, including autoregressive policies like OpenVLA [5], and π_0 -FAST [14], as well as diffusion/flow-based policies such as π_0 [6] and Octo [3]. Our results consistently show that leveraging self-certainty for test-time scaling significantly improves performance over the respective baselines: greedy decoding for autoregressive models and standard single-pass inference for diffusion-based models.

2 Related Work

Vision-Language-Action models. Recent advancements in machine learning have spurred the development of robotic foundation models [2, 3, 15, 16, 17, 5, 18], with a significant focus on VLA models [4, 5, 6, 14, 19, 20, 21, 8, 13, 7, 22, 23, 24, 25]. These models typically leverage the rich world knowledge of pre-trained Vision-Language Models (VLMs) by fine-tuning them on large-scale, diverse robot datasets [26, 27, 28, 29, 30, 31, 32]. This paradigm has produced generalist policies capable of impressive generalization across a wide variety of manipulation tasks [7, 8]. A key architectural distinction among VLAs lies in their action representation. One approach, mirroring the success of LLMs, discretizes continuous robot actions into a sequence of tokens and predicts them auto-regressively [4, 5, 14]. In contrast, another line of work preserves the continuous nature of actions, employing diffusion [15, 6, 7] or flow-matching models [6, 8] for action generation. Our proposed method is designed to be agnostic to this choice and can be readily applied to both autoregressive and diffusion-based VLA policies.

Test-time scaling via sample-and-verify. The stochastic nature of generative models permits a powerful inference-time strategy known as the sample-and-verify approach, or Best-of-N selection [33, 34]. This paradigm involves generating multiple candidate outputs and using a criterion to select the best one. Typically, this selection is guided by an externally trained verifier or a reward model, which evaluates the quality of each candidate. This has proven to be an effective test-time scaling framework, where increasing computation at inference can be more beneficial than scaling model parameters during training [35]. In the robotics domain, this paradigm has been successfully applied to VLAs. For instance, RoboMonkey [12] and V-GPS [11] both employ learned verifiers or value functions to select the best action chunk from multiple proposals.

However, training such verifiers is computationally expensive and introduces practical challenges. To address this, recent work by Kang et al. [1] introduced a verifier-free alternative for LLMs. They

proposed *self-certainty*, a metric derived from the model’s own output probabilities, to rank reasoning chains and select high-quality responses without the need for an external reward model. Our work builds directly upon this insight, adapting and extending the principle of self-certainty from the discrete domain of language to the continuous action spaces of diffusion-based VLA models. Other test-time methods like BID [36] and PTP [37], while also being verifier-free like ours, focus on a different objective. Their primary aim is to enforce temporal consistency across action chunks rather than to identify the globally optimal action from a set of diverse candidates.

Confidence calibration in VLAs. A foundational study by Zollo & Zemel [38] provides the first systematic analysis of confidence calibration in VLAs. Their work establishes a baseline confidence metric for models like OpenVLA and reveals that higher task success rates are strongly correlated with better calibration (i.e., lower calibration error). This discovery provides the bedrock for our approach. While their work focuses on measuring and improving calibration, we treat the VLA’s confidence score as an actionable signal for test-time scaling. Our method is a natural extension of their findings, operating on the premise that a VLA’s confidence is a reliable enough indicator of success to facilitate the selection of the best action from multiple candidates.

3 Preliminaries

Problem formulation. We frame robot control as a Markov Decision Process (MDP), where a policy $\pi(\mathbf{A}_t|o_t)$ takes an observation o_t (including images and language) and outputs a continuous action chunk $\mathbf{A}_t \in \mathbb{R}^{Nd}$, representing N future d -dimensional actions.

Autoregressive policies. These policies first tokenize the continuous action chunk $\mathbf{A}_t$ into a sequence of M discrete tokens $\mathbf{y}$. The probability of this sequence is then modeled autoregressively:

$$p(\mathbf{y}|o_t) = \prod_{i=1}^M p(y_i|y_{<i}, o_t). \quad (1)$$

Our method leverages the per-token probability distribution, $p(\cdot|y_{<i}, o_t)$, which is derived from the model’s logits via a softmax function.

Diffusion-based policies. These policies learn a denoiser network ϵ_θ to reverse a noising process. Inference starts with random noise $\mathbf{A}^T$ and iteratively applies the denoiser for T times to recover a clean action chunk $\mathbf{A}^0$. Crucially for our method, the predicted clean action $A_{0,\text{pred},t}$ can be estimated from any noisy state $\mathbf{A}^t$ [39] using:

$$\mathbf{A}^0 \approx A_{0,\text{pred},t} = \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{A}^t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(\mathbf{A}^t, t)), \quad (2)$$

where $\bar{\alpha}_t$ is a noise schedule parameter.

Flow-based policies. Models like π_0 learn a vector field v_θ to generate actions from noise by solving an ordinary differential equation (ODE). Inference involves integrating this field, typically with a numerical solver like forward Euler:

$$\mathbf{A}^{\tau-\delta} = \mathbf{A}^\tau + \delta \cdot v_\theta(\mathbf{A}^\tau, o_t), \quad (3)$$

where δ is the integration step size. This is repeated from $\tau_T = 1$ to $\tau_0 = 0$ to predict the clean action chunk $\mathbf{A}^0$ from noise $\mathbf{A}^T$ with T steps (each with the same integration step size).

4 Method

Our goal is to develop a test-time scaling method for VLA models that enhances performance without the need for an externally trained verifier. We achieve this by defining a *self-certainty* metric that quantifies the model’s intrinsic confidence in its own generated actions. A key challenge is that VLAs employ different action generation paradigms. We therefore first adapt the original self-certainty concept [1] for autoregressive VLAs, and then propose a novel formulation for the continuous action spaces of diffusion and flow-based models. Finally, to overcome the inapplicability of

discrete voting methods [1] to the continuous action spaces of VLAs, we introduce a novel selection mechanism that combines self-certainty scores with similarity-based agreement for robust action selection.

4.1 Self-Certainty for Autoregressive VLAs

For autoregressive VLAs such as OpenVLA [5], which generate actions as a sequence of discrete tokens, we can directly apply the distributional confidence metrics proposed by Kang et al. [1]. This work found that metrics evaluating the entire token probability distribution are more robust than those based solely on the probability of the sampled token. We investigate the two most effective metrics from their study:

KL-divergence-based self-certainty. This metric is defined as the average KL-divergence between the model’s predicted token distribution $p(\cdot|\cdot)$ and a uniform distribution $\mathcal{U}$ over the vocabulary. For an token sequence y , the score is:

$$S_{\text{cert}}(A^t) = \frac{1}{N} \sum_{i=1}^N \text{KL}(\mathcal{U} \| p(\cdot|o_t, y_{<i})). \quad (4)$$

A higher score indicates the predicted distribution is more peaked and less uniform, signifying higher confidence.

Entropy-based self-certainty. As an alternative, we measure the negative entropy of the predicted distribution, which also quantifies its peakedness:

$$S_{\text{cert}}(A^t) = \frac{1}{N} \sum_{i=1}^N \sum_{k \in \mathcal{V}} p(k|o_t, A_{<i}^t) \log p(k|o_t, A_{<i}^t), \quad (5)$$

where $\mathcal{V}$ is the token vocabulary.

4.2 Self-Certainty for Continuous Action Models

A core contribution of our work is to extend the principle of self-certainty to VLAs that generate continuous actions, such as those based on diffusion or flow-matching, which lack a discrete vocabulary.

4.2.1 Diffusion-Based Policies

For diffusion policies like Octo [3], which iteratively denoise a random vector to produce an action, our intuition is that a confident model will converge on its final prediction early in the denoising process. We can quantify this “denoising stability” as follows:

1. During each step t of the T -step denoising process, the model predicts the noise $\epsilon_\theta(A^t, t)$ from the noisy action A^t . We can then estimate the model’s current prediction for the final, clean action $A^{0\text{-pred},t}$ using Eq. (2).
2. We measure the average cosine similarity between consecutive predictions of the final action in the last K steps of the denoising trajectory, and the self-certainty score is the average of the cosine similarities:

$$S_{\text{cert}}(A) = \frac{1}{K} \sum_{t=0}^{K-1} \cos(A^{0\text{-pred},t}, A^{0\text{-pred},(t+1)}). \quad (6)$$

A higher score signifies a more stable denoising process and thus higher confidence.

4.2.2 Flow-Based Policies

For flow-based policies like π_0 [6], which generate actions by integrating a learned vector field v_θ , our intuition is that a confident prediction corresponds to a smooth and direct integration path. The vector field should consistently point toward the final destination. We measure this as follows:

1. First, we generate a complete action chunk A^0 by integrating from an initial noise vector A^1 over all steps using Eq (3).
2. Then, for the last K steps of the integration trajectory (from τ_K to τ_0), we compute the cosine similarity between the learned vector field $v_\theta(A^{\tau_j}, o_t)$ and the direct path to the final action, $(A^0 - A^{\tau_j})$, and the self-certainty score is the average of these cosine similarities:

$$S_{\text{cert}}(A) = \frac{1}{K} \sum_{j=1}^K \cos(v_\theta(A^{\tau_j}, o_t), A^0 - A^{\tau_j}). \quad (7)$$

A score closer to 1 indicates a more direct, confident trajectory.

4.3 Best-of-N Selection via Combined Certainty and Similarity

Simply selecting the single candidate with the highest self-certainty can be sensitive to outliers with spuriously high confidence [1]. To create a more robust selection mechanism, we combine the model’s intrinsic confidence with a measure of consensus among the sampled candidates.

For a set of N sampled action chunks $\{A_1, A_2, \dots, A_N\}$, we select the best one, A^* , by maximizing a combined score:

$$A^* = \arg \max_{A_i} S_{\text{final}}(A_i). \quad (8)$$

The final score for each candidate A_i is a weighted sum of its self-certainty score and its agreement with other candidates:

$$S_{\text{final}}(A_i) = \lambda \cdot S_{\text{cert}}(A_i) + (1 - \lambda) \cdot S_{\text{sim}}(A_i), \quad (9)$$

where:

- $S_{\text{cert}}(A_i)$ is the self-certainty score of action chunk A_i , calculated using the methods described in Section 4.1 or 4.2.
- $S_{\text{sim}}(A_i) = \frac{1}{N-1} \sum_{j \neq i} \text{similarity}(A_i, A_j)$ is the agreement score, defined as the average pairwise similarity of A_i to all other candidates. Similarity can be measured by negative MSE or cosine similarity (we use MSE in this paper).
- λ is a hyperparameter that balances the two components.

This approach allows us to favor actions that are not only deemed confident by the model’s internal metrics but are also part of a consistent group of solutions found across multiple samples.

5 Experiments

We conduct simulation and real-world experiments to validate our proposed test-time scaling framework. Our evaluation aims to answer the following questions: (1) How do the core components of our method—namely the number of samples (N), the choice of self-certainty metric, the number of last steps in diffusion (K), and the inclusion of a similarity score—affect performance? (2) Does our method provide significant performance gains over standard inference baselines for a variety of state-of-the-art VLA models on established benchmarks?

5.1 Experimental Setup

Benchmarks We evaluate our method on two distinct simulation benchmarks to test its versatility across different tasks and environments:

- **SIMPLER**: A suite of simulated environments designed for evaluating real-world robot manipulation policies. We evaluate on four Google Robot tasks (we combine drawer tasks into a single one): *Pick Coke Can*, *Move Near*, and *Drawer* (Open/Close).

- **LIBERO:** A benchmark that features a diverse set of manipulation skills with human-teleoperated demonstrations. Following π_0 -FAST, [14], we finetune generalist robot policies including π_0 [6] and π_0 -FAST on the combination of the four task subsets—LIBERO-Spatial, LIBERO-Object, LIBERO-Goal, and LIBERO-10—and evaluate on each of them.

Models We apply our method to four pretrained VLA models, covering different architectures, sizes, and action generation paradigms:

- **Autoregressive Policies:** OpenVLA [5], a 7B parameter VLA that discretizes action dimensions into action bins, and π_0 -FAST [14], a 3B state-of-the-art generalist robot policy which uses learned action tokenization, with architecture similar to π_0 [6].
- **Diffusion/Flow-Based Policies:** Octo [3], a generalist policy that uses a diffusion head to predict continuous actions, and π_0 [6], which employs a flow-based model for action generation.

Evaluation Metric and Baselines The primary evaluation metric across all experiments is the task success rate (%). Our approach is compared against the standard inference procedure for each model class: greedy decoding for autoregressive policies like OpenVLA, and a standard single pass for diffusion or flow-based policies. By default, we set λ to 0.5 and token output temperature for autoregressive policies to 0.5.

5.2 Ablation Studies

We first conduct ablation studies to determine the optimal configuration for our method.

5.2.1 Core Component Analysis

Figure 1 shows the impact of sample count (N) and different selection strategies on the average task success rate for OpenVLA on the SIMPLER benchmark. The results clearly indicate that combining self-certainty and similarity with a KL-Divergence metric at $N=8$ yields the best performance. We therefore use this configuration for all subsequent benchmark evaluations.

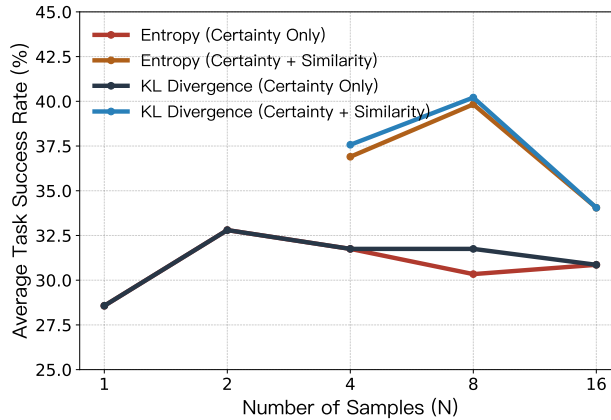


Figure 1: Ablation study on OpenVLA on the SIMPLER benchmark. The plot compares the average task success rate of strategies using certainty only versus certainty combined with similarity across different numbers of samples (N). Note: we cannot implement similarities (agreement score) when $N \leq 2$.

5.2.2 Hyperparameter Sensitivity

We also study the effect of the number of K —the last K steps of the denoising trajectory that used to calculate self-certainty—on model performance. Figures 2 show that a moderate number of $K = 10$

leads to best performance. Thus we adopt this in the following experiments. For flow-based policy π_0 , since the total inference steps is small (10), we just set $K = 10$.

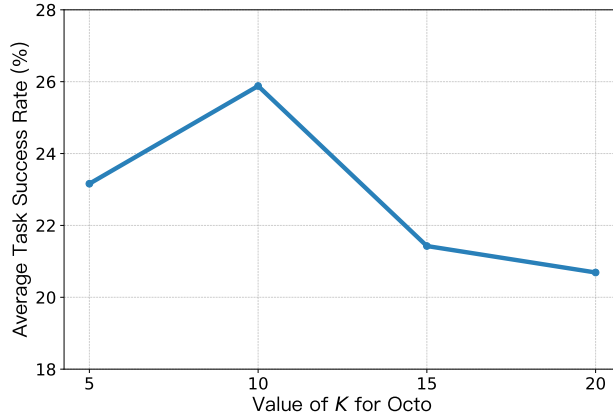


Figure 2: The influence of the number of last steps K on SIMPLER performance of diffusion-based policies (Octo).

5.3 Benchmark Evaluation

We evaluated the selected models on the SIMPLER and LIBERO benchmarks using the optimal settings determined from the ablation studies (KL-Divergence, N=8, Certainty + Similarity, K=10).

5.3.1 SIMPLER Benchmark

Table 1 summarizes the results for OpenVLA and Octo on the SIMPLER benchmark. Our method significantly improves performance for both models on several tasks. For example, the success rate of OpenVLA on the *Drawer (Open/Close)* task increased from 33.33% to 48.15%. This demonstrates that our method can be useful for both autoregressive and diffusion-based policies. However, our preliminary results on Bridge tasks, where the baseline success rate of OpenVLA was zero, our method did not yield any improvement. This suggests that our test-time scaling technique can enhance a model’s existing, albeit unreliable, capabilities, but it cannot create new abilities that the model completely lacks.

Table 1: Task success rates (%) on the SIMPLER benchmark. Rows denote the model and columns denote the task. For our method, the absolute improvement over the baseline is shown in parentheses.

Model	Pick Coke	Move Near	Drawer	Average
OpenVLA	12.00	50.00	33.33	31.78
OpenVLA + Ours	16.00 (+4.00)	56.67 (+6.67)	48.15 (+14.82)	40.28 (+8.50)
Octo	29.33	6.67	31.48	22.49
Octo + Ours	41.33 (+12.00)	6.67 (+0.00)	31.48 (+0.00)	26.49 (+4.00)

5.3.2 LIBERO Benchmark

Table 2 shows the results for the finetuned π_0 and π_0 -FAST models on the LIBERO benchmark. Our method consistently improved the performance of both state-of-the-art models. For the π_0 model, the average success rate increased from 89.65% to 91.85%. For π_0 -FAST, the average success rate also increased from 80.35% to 81.70%. These results demonstrate that our method can enhance strong policies, and can be applied to flow-based models as well.

Table 2: Task success rates (%) on the LIBERO benchmark. Rows denote the model configuration and columns denote the task suite. For our method, the absolute improvement over the baseline is shown in parentheses.

Model	LIBERO-Spatial	LIBERO-Object	LIBERO-Goal	LIBERO-10	Average
π_0	92.00	97.60	91.00	78.00	89.65
π_0 + Ours	97.00 (+5.00)	99.20 (+1.60)	92.60 (+1.60)	78.60 (+0.60)	91.85 (+2.20)
π_0 -FAST	92.20	95.80	80.80	52.60	80.35
π_0 -FAST + Ours	95.60 (+3.40)	96.80 (+1.00)	80.80 (+0.00)	53.60 (+1.00)	81.70 (+1.35)

5.4 Real-World Validation

To validate the effectiveness of our method in the physical world, we conducted a real-robot experiment with an AgileX PIPER robotic arm. We use a third-person camera and a wrist camera, as shown in Figure 3. We used the pretrained π_0 -fast model and performed full fine-tuning on 30 demonstration trajectories for the “pick the snack into the tray” task. In 10 test trials, the baseline method (greedy decoding) succeeded once (10% success rate), whereas our method succeeded 3 times (30% success rate). This result confirms that the performance gains observed in simulation can be successfully transferred to real-world manipulation tasks, validating the practical applicability of our test-time scaling framework.

6 Conclusion

This paper introduces a “plug-and-play”, verifier-free framework to improve the test-time performance of Vision-Language-Action (VLA) models by adapting the principle of self-certainty. We propose novel self-certainty metrics for the continuous action spaces of diffusion and flow-based policies, overcoming the limitations of prior work focused on discrete domains. By sampling multiple actions and selecting the one with the optimal combination of a self-certainty score and a similarity-based agreement score, our method trades increased computation for enhanced reliability. Extensive experiments in simulation and the real world demonstrate that this approach significantly improves task success rates across diverse VLA architectures, including autoregressive, diffusion, and flow-based models, establishing it as a general strategy for scaling VLA performance at test time.

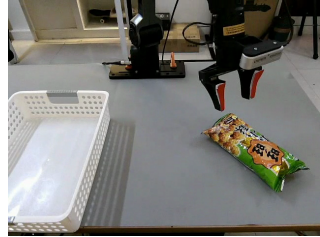


Figure 3: Real-world experiment setup.

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